

SD-WAN Configuration Documentation

HQ and Branch FortiGate Deployment

1. Network Topology Overview

The network consists of two sites:

Headquarters (HQ)

WAN Path

- Two routers provide dual WAN paths between sites:
 - **R1:** 10.10.10.1
 - **R2:** 10.20.20.1
- Both HQ and Branch FortiGate units connect to an IOU1 switch via VLAN trunk links.

```
R2
R2(config-if)#ip address 10.10.2.1 255.255.255.252
R2(config-if)#no sh
R2(config-if)#
*Nov 28 10:07:01.319: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Nov 28 10:07:02.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#no ip address
R2(config-if)#exit
R2(config)#exit
R2#sh
*Nov 28 10:18:20.883: %SYS-5-CONFIG_I: Configured from console by console
R2#show sys int
^
% Invalid input detected at '^' marker.

R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 10.20.20.1 255.255.255.252
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
```

```
R1
et0/0, changed state to up
R1(config-if)#exit
R1(config)#int f0/1
^
% Invalid input detected at '^' marker.

R1(config)#int f0/0
R1(config-if)#no ip address
R1(config-if)#exit
R1(config)#exit
R1#
*Nov 28 10:19:07.403: %SYS-5-CONFIG_I: Configured from console by console
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.252
R1(config-if)#wr memory
^
% Invalid input detected at '^' marker.

R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
```

2. HQ FortiGate Configuration

2.1 WAN Interface Sub-Interfaces

Two VLAN-based WAN interfaces were created for the dual routers.

GUI Path:

Network → Interfaces → Create New → Interface

Parameter	WAN_R1	WAN_R2
Type	VLAN	VLAN
Parent Interface	port5	port5
VLAN ID	10	20
IP	10.10.10.2/30	10.20.20.2/30
Role	WAN	WAN
Access	PING	PING

CLI Equivalent

```
config system interface
  edit WAN_R1
    set vdom root
    set ip 10.10.10.2 255.255.255.252
    set allowaccess ping
    set role wan
    set interface port5
    set vlanid 10
  next
  edit WAN_R2
    set vdom root
    set ip 10.20.20.2 255.255.255.252
    set allowaccess ping
    set role wan
    set interface port5
    set vlanid 20
  next
end
```

Connectivity Test

execute ping 10.10.10.1 → Success

execute ping 10.20.20.1 → Success

CLI Console (1)

```
FortiGate-1 # execute ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1): 56 data bytes
64 bytes from 10.10.10.1: icmp_seq=0 ttl=255 time=536.8 ms
64 bytes from 10.10.10.1: icmp_seq=1 ttl=255 time=20.6 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=255 time=14.6 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=255 time=15.9 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=255 time=13.4 ms

--- 10.10.10.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 13.4/120.2/536.8 ms

FortiGate-1 # execute ping 10.20.20.1
PING 10.20.20.1 (10.20.20.1): 56 data bytes
64 bytes from 10.20.20.1: icmp_seq=0 ttl=255 time=60.8 ms
64 bytes from 10.20.20.1: icmp_seq=1 ttl=255 time=13.7 ms
64 bytes from 10.20.20.1: icmp_seq=2 ttl=255 time=20.2 ms
64 bytes from 10.20.20.1: icmp_seq=3 ttl=255 time=22.1 ms
64 bytes from 10.20.20.1: icmp_seq=4 ttl=255 time=13.8 ms

--- 10.20.20.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 13.7/26.1/60.8 ms

FortiGate-1 #
```

2.2 SD-WAN Zone and Members

GUI Path:

Network → SD-WAN → SD-WAN Zones

Zone Name: HQ_SD-WAN

Members Added:

Member	Interfac	Gateway
1	WAN_R1	

		10.10.10.1	
2	WAN_R2	10.20.20.1	
	 SDWAN_ZONE		
	•  port5	192.168.1.1	0
	•  port2	192.168.1.2	0
	•  WAN_R1	10.10.10.1	10
	•  WAB_R2	10.20.20.1	20

2.3 Performance SLAs

GUI Path:

Network → SD-WAN → Performance SLAs

SLA Name	Monitored Interface	Probe Target	Metrics
SLA_R1	WAN_R1	8.8.8.8	Latency, Jitter, Packet Loss
SLA_R2	WAN_R2	1.1.1.1	Latency, Jitter, Packet Loss

Thresholds:

- Latency: **50 ms**
- Jitter: **20 ms**
- Packet Loss: **5%**

SLA status verified as UP.

Edit Performance SLA

Name

SLA_R2

Probe mode

Active

Passive

Prefer Passive

Protocol

Ping

HTTP

DNS

Server

1.1.1.1

+

Participants

All SD-WAN Members

Specify

WAB_R2

+

SLA Target

Latency threshold

50

ms

Jitter threshold

20

ms

Packet Loss threshold

5

%

Link Status

Check interval

500

ms

Failures before inactive

5

Restore link after

5

check(s)

Actions when Inactive

Update static route

OK

Cancel

+ Create New

Edit

Delete

Search

Name	Detect Server	Packet Loss	Latency	Jitter	Failure Threshold	Recovery Thr
Default_DNS	96.45.45.45 96.45.46.46 (System DNS)				5	10
Default_FortiGuard	http://fortiguard.com/				5	10
Default_Gmail	gmail.com				5	10
Default_Google Search	http://www.google.com/				5	10
Default_Office_365	http://www.office.com/				5	10
SLA_BR_R1	10.10.10.1	WAN_R1_BR: 18.00%	WAN_R1_BR: 18.84ms	WAN_R1_BR: 7.95ms	5	5
SLA_BR_R2	10.20.20.1	WAN_R2_BR: 74.00%	WAN_R2_BR: 19.40ms	WAN_R2_BR: 8.35ms	5	5

Edit Performance SLA

Name

SLA_R1

Probe mode

Active

Passive

Prefer Passive

Protocol

Ping

HTTP

DNS

Server

9.9.9.9

Participants

All SD-WAN Members

Specify

WAN_R1

SLA Target

Latency threshold

50

ms

Jitter threshold

20

ms

Packet Loss threshold

5

%

Link Status

Check interval

500

ms

Failures before inactive

5

Restore link after

5

check(s)

Actions when Inactive

Update static route

OK

Cancel

2.4 SD-WAN Rule

GUI Path:

Network → SD-WAN → SD-WAN Rules

Field	Value
Rule Name	HQ-BR-VPN
Source	all
Destination	Branch remote subnets (HQ-To-BR objects)
Strategy	Best Quality
Preferred Interfaces	WAN_R1, WAN_R2
Measured SLA	SLA_R1

Quality Criteria

Latency

Priority Rule

Outgoing Interfaces

Select a strategy for how outgoing interfaces will be chosen.

☐ Manual
Manually assign outgoing interfaces.

☒ **Best Quality**
The interface with the best measured performance is selected.

☐ Lowest Cost (SLA)
The interface that meets SLA targets is selected. When there is a tie, the interface with the lowest assigned cost is selected.

☐ Maximize Bandwidth (SLA)
Traffic is load balanced among interfaces that meet SLA targets.

Interface preference

WAN_R1_BR

×

WAN_R2_BR

×

+

Zone preference

+

Measured SLA

SLA_BR_R1

▼

Quality criteria

Latency

▼

Forward DSCP

☐

Reverse DSCP

☐

Status

Enable

Disable

OK

Cancel

3. Branch FortiGate Configuration

Branch configuration mirrors HQ, using _BR suffixes.

3.1 WAN Sub-Interfaces

Parameter	WAN_R1_BR	WAN_R2_BR
Type	VLAN	VLAN
Parent Interface	port6	port6
VLAN ID	10	20

IP	10.10.10.x/30	10.20.20.x/30
Role	WAN	WAN
Access	PING	PING

CLI

```
config system interface
  edit WAN_R1_BR
    set ip 10.10.10.x 255.255.255.252
    set vlanid 10
    set allowaccess ping
    set interface port6
  next
  edit WAN_R2_BR
    set ip 10.20.20.x 255.255.255.252
    set vlanid 20
    set allowaccess ping
    set interface port6
  next
end
```

192.168.110.132 - PuTTY

```
FortiGate_1 # conf sys int
FortiGate_1 (interface) # edit WAN_R2_BR
FortiGate_1 (WAN_R2_BR) # set ip 10.20.20.2 255.255.255.252
FortiGate_1 (WAN_R2_BR) # set interface port6
FortiGate_1 (WAN_R2_BR) # set vlanid 20
FortiGate_1 (WAN_R2_BR) # set role wan
FortiGate_1 (WAN_R2_BR) # set allowaccess ping
FortiGate_1 (WAN_R2_BR) # next
FortiGate_1 (interface) # do wr
Unknown action 0
FortiGate_1 (interface) # Timeout
FortiGate_1 login: 
```

3.2 SD-WAN Zone

Zone Name: BR_SD-WAN

Members:

Interface		Gateway			
WAN_R1_BR		10.10.10.1			
WAN_R2_BR		10.20.20.1			

	BR_SD-WAN				
	WAN_R1_BR	10.10.10.1	0	1.51 kbps	
	WAN_R2_BR	10.20.20.1	0	417 bps	

3.3 Performance SLAs

SLA Name	Interface	Target
----------	-----------	--------

SLA_BR_R1	WAN_R1_BR	10.10.10.1
SLA_BR_R2	WAN_R2_BR	10.20.20.1

Thresholds identical to HQ.



Edit Performance SLA

Name	SLA_BR_R1
Probe mode	Active Passive Prefer Passive
Protocol	Ping HTTP DNS
Server	10.10.10.1
Participants	All SD-WAN Members Specify WAN_R1_BR

SLA Target

Latency threshold	50	ms
Jitter threshold	20	ms
Packet Loss threshold	5	%

Link Status

Check interval	500	ms
Failures before inactive	5	
Restore link after	5	check(s)

Actions when Inactive

Update static route	
---------------------	--

Edit Performance SLA

Name

SLA_BR_R2

Probe mode ⓘ

Active

Passive

Prefer Passive

Protocol

Ping

HTTP

DNS

Server

10.20.20.1

+

Participants

All SD-WAN Members

Specify

WAN_R2_BR

×

+

SLA Target ☒

Latency threshold ☒

50

ms

Jitter threshold ☒

20

ms

Packet Loss threshold ☒

5

%

Link Status

Check interval

500

ms

Failures before inactive ⓘ

5

Restore link after ⓘ

5

check(s)

Actions when Inactive

Update static route ⓘ

☒

OK

Cancel

3.4 SD-WAN Rule

Field	Value
Name	BR-TO-HQ-VPN
Source	all
Destination	HQ remote subnets
Strategy	Best Quality

Preferred
Interfaces WAN_R1_BR, WAN_R2_BR

Measured SLA SLA_BR_R1

Criteria Latency

SD-WAN Zones SD-WAN Rules Performance SLAs						
+ Create New Edit Clone Delete Search Q						
ID	Name	Source	Destination	Criteria	Members	Hit Count
IPv4 1						
1	BR-TO-HQ-VPN	all	BR-To-HQ_remote_subnet_1 BR-To-HQ_remote_subnet_2 BR-To-HQ_remote_subnet_3 BR-To-HQ_remote_subnet_4 BR-To-HQ_remote_subnet_5	Latency	WAN_R1_BR WAN_R2_BR	0

4. Status Summary

Component	HQ	Branch
WAN Interfaces	OK	OK
Gateway	OK	OK
Reachability		
SD-WAN Zone	Configured	Configured
SLA Monitoring	Active	Active
Rules	Applied	Applied